CS1910 Computer Science I • Input and Output **Partner workbook**

Sender and Receiver

In lecture with a partner • about 25 minutes

Name: _____ **Partner:** _____

This PDF can be filled in: click in an answer space to type, or print it and write. Draw diagrams with your PDF viewer's pen or markup tool, or on paper.

What this workbook practises

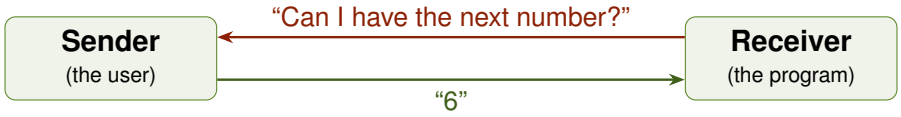
Before writing Python, you will act out a program with your partner. One of you plays the user, the other plays the program. Then you will use computational thinking to turn what you did into an algorithm, and finally write a short program that reads input from the user.

1 Act it out: the largest number so far

about 8 minutes

The problem

Simulate a program that reads numbers from the user (forever, but we will stop after 6) and, after each number, reports the **largest number seen so far**. Example: `5, 6, 3, 2, 6, 8`.



1. The **sender** secretly writes down 6 numbers.
2. One at a time, the **receiver** asks "Can I have the next number?" and the sender says it.
3. Repeat 6 times. Then the sender asks "What was the highest number I gave you?" and the receiver answers.

1. Receiver: as you play, record what you hear and what you are keeping track of.

Number #	1	2	3	4	5	6
Number heard						
Largest so far						

2. Receiver: did you need to remember *all* of the numbers to answer the final question? What did you actually keep in your head?

2 Think it through

about 8 minutes

3. What are we trying to achieve? What do we know? What information is missing?

4. **Decomposition:** label each step of the activity as *input*, *output* or *processing*.

Step	Input, output or processing?
Receiver asks “Can I have the next number?”	
Sender says the next number	
Receiver compares it with the largest so far	
Receiver reports the highest number	

5. **Patterns:** what repeats? Have you seen anything that compares maximum values?

6. What simple or unusual cases should the program handle?

7. **Algorithm:** write pseudocode for the program (no Python yet).

3 Write the code: average of three numbers

about 8 minutes

Problem

Write a program that asks the user to enter 3 floating-point numbers, one after the other (not all on the same line), and then prints their average. Use the `+=` (add and assign) operator in your solution.

8. Write the program.

9. What does `input` return? What goes wrong if you leave out `float(...)`?

Before you leave

about 1 minutes

- [ ] I can label the steps of a program as input, output or processing.
- [ ] I can keep a running value (largest so far, total) instead of storing every input.
- [ ] I can convert input to a number before doing arithmetic.